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ISO 9000 and other metastandards: Principles for management practice?

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Executive Overview

Managers around the world are witnessing the spread of standards for evaluating managerial practices. Recent examples include the International Organization for Standardization (ISO) 9000, the US Federal Sentencing Guidelines, recent regulations of the Occupational Safety and Health Administration (OSHA), environmental management standards, and the Committee of Sponsoring Organizations of the Treadway Commission (COSO) Framework internal controls. These metastandards constitute a new management technology that may bring meaningful standardization to general management practice for the first time in history. But do standards reflect alchemy or science? Can best-laid standards, like best-laid plans, go awry?

Introduction

For three decades, managers and scholars shared the belief that organizations need customized management systems and strategies. In the late 1960s, landmark studies conclusively showed that management practice was too diverse for simplistic solutions. As Philip K. Howard notes, this view is buttressed by government's many failures with regulatory standards that impinge on management practice:

> Our regulatory system has become an instruction manual. It tells us and bureaucrats exactly what to do and how to do it. Detailed rule after detailed rule addresses every eventuality, or at least every situation lawmakers and bureaucrats can think of. Is it a coincidence that almost every encounter with government is an exercise in frustration?

Nonetheless, a few legal, regulatory, and standards-making bodies continue to seek general principles. In the late 1980s, motivated by threats to quality, health and safety, they began to make major breakthroughs. Management system standards emerged in fields as diverse as product quality (the ISO 9000 family of standards), financial controls (the COSO Framework), and white collar crime prevention (the US Federal Sentencing Guidelines for managerial due diligence). These standards rapidly attracted support from powerful stakeholders. These stakeholders have forced thousands of organizations to comply. Now managers and scholars face urgent questions: How do these new standards work? What do they demand? When and how should they be used? Above all, what are the potential long-term side-effects?

This discussion of these questions begins by defining a metastandard, the technological breakthrough that makes management standardization possible. To see how metastandards are applied, we look at ISO 9001, the most advanced specimen to date. We then consider the growing diversity of metastandards

and the forces that are causing them to spread so quickly. Finally, we pose scenarios for a time when metastandards are routinely applied to large numbers of organizations. These scenarios should stimulate a critical examination of a management revolution in the making.

What is a Metastandard?

Over the past century, governments and organizations have produced multitudes of standards, laws, and regulations. With so much experience, it seems absurd to suggest that we might now be witnessing the birth of a new class of standards. The suggestion becomes plausible, however, when one observes that the standardization universe has always had a gaping void. To help visualize this gap, Figure 1 divides standardization efforts across two dimensions: a) the stage of process that is being standardized, and b) the conceptual level at which the standard is written.

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Managers are most familiar with input and output standards, which are common because they deal directly with things that most organizations want to standardize, such as product specifications, job qualifications, environmental limits, and customer service goals, and because they can be easily worded for any level of generality. The text can be narrow and precise (e.g., test specifications) or abstract and general (e.g., the corporate mission statement).

Input and output standards, however, are poor predictors of future outputs. A product sample may pass a lab test with flying colors, but the unit has already been made. To expect good output in the future, one must have confidence in the management process. A competent management process will presumably do whatever is necessary to ensure future quality. With a poor management process, anything could happen.

If management tasks are very specific (e.g., specific test procedures or job descriptions), management standards are easily written. As tasks are assembled into larger management systems, however, the standards become harder to

Level	Stage: Inputs (Suppliers →)	Management Process	Outputs (→ Stakeholders)
metastandard *(rules for designing systems of items)*	performance criteria for inputs *(building code)*	***management system "metastandard"***	performance criteria for outputs *(goals for customer satisfaction)*
system standard *(description of system of items)*	required system of inputs *(fire safety plan)*	design of a specific management system *(system for responding to customer complaints)*	required system of outputs *(product specification)*
standard *(description of an item)*	specific input requirement *(specification for safety door)*	detailed task description *(procedure for answering the telephone)*	specific output requirement *(finishing tolerances)*

Figure 1 - Categories of Standardization

write. Systems have complex interactions that make them hard to codify. In the past, standards writers responded by applying a specific system design universally. Unfortunately, this heavy-handed approach was acceptable only when health, safety, or major financial issues were at stake.

If instructions are made sufficiently abstract, however, they eventually cross an important threshold. Rather than writing detailed instruction manuals, standards writers can create lists of design rules to guide the creation of entire classes of management systems. Since systems theorists use the term *metasystem* for lists of this type, it follows that this type of management standard should be referred to as a metastandard.[2]

The key breakthrough in standardizing management practice came when standards-writers began to craft metastandards. The significance of this became evident when the first important example emerged to affect tens of thousands of companies around the world. ISO 9001, the metastandard document in the ISO 9000 family of quality standards, is arguably the most influential single metastandard so far. Figure 2 shows how it is typically applied.[3]

ISO 9001 ignores inputs or outputs. It assumes that the supplier and the customer will negotiate these in the sales contract. Instead, ISO 9001 defines a quality management system that can protect virtually any type of product or service. The metastandard lists twenty essential quality management subsystems. To comply, a supplier must effectively implement *every* subsystem in Table 1.

The clauses in ISO 9001 specify what subsystems are required, rather than how they should operate. ISO 9001 implicitly acknowledges that, while different management systems should have the same set of subsystems, those subsystems may operate differently from one company to the next.

The clauses in ISO 9001 specify what subsystems are required, rather than how they should operate. ISO 9001 implicitly acknowledges that, while different management systems should have the same set of subsystems, those subsystems may operate differently from one company to the next. This distinction is preserved throughout the standard. For example, clause 4.3 (contract review) requires suppliers to review all contracts to ensure that both parties understand and agree to the provisions of the sale and to verify that the supplier can meet its promises before it commits to the sale:

> Before submission of a tender, or at the acceptance of a contract or order (statement of requirement), the tender, contract, or order shall be reviewed by the supplier to ensure that: a) the requirements are adequately defined and documented; where no written statement of requirement is available for an order received by verbal means, the supplier shall ensure that the order requirements are agreed before their acceptance; b) any differences between the contract or accepted order requirements and those in the tender are resolved; c) the supplier has the capability to meet the contract or accepted order requirements.

To satisfy clause 4.3, the supplier must do more than admonish its salespersons to be honest and careful. The supplier must create a closed-loop management subsystem to review every transaction. The design of the subsystem is left to the supplier. One supplier might give cellular notebooks to each salesperson to confirm orders directly as they are taken. Another might publish a daily list of in-stock items, and require that other orders be confirmed by telephone. The owner-salesperson in a small business might get by with a well-organized

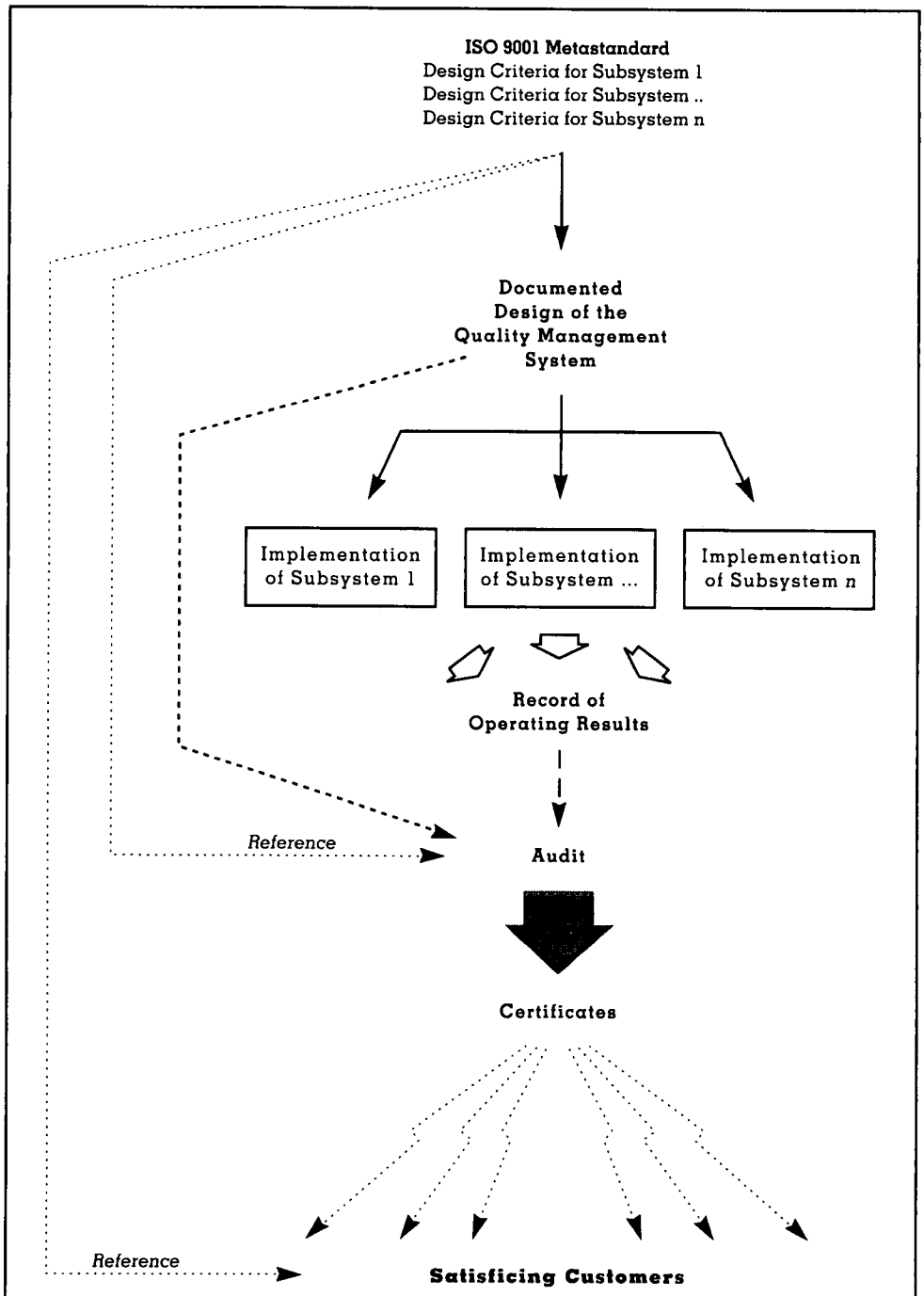

Figure 2 - Structure of a Metastandard Compliance System

ledger summarizing current orders, inventory and production schedules. If followed faithfully, each system would be equally valid.

Once the organization designs the management system and implements the subsystems, an auditor is engaged to perform a quality audit. The auditor begins by comparing the design of the quality system to the clauses in the standard. If the design is complete, it is approved in principle. The auditor then looks for evidence that the organization is effectively implementing each

Table 1
Paraphrase of the ISO 9001 Metastandard[4]

Clause	Required Management Subsystem
4.1	A system of management for the quality system, including a policy, organization, assigned responsibilities, and a review mechanism that involves senior management.
4.2	A documented plan for the quality system.
4.3	A system to ensure that customer and supplier clearly understand and agree to their contract.
4.4	A system to control and verify the design to ensure that it meets specified requirements.
4.5	A system to prevent errors due to inadequate or out-of-date documentation.
4.6	A system to ensure deliberate purchasing decisions and the use of qualified suppliers.
4.7	A system to safeguard any materials that are entrusted to the supplier by the customer.
4.8	A system to trace units of product through production (if required by the sales contract).
4.9	A system to ensure that the produce is made in a known, planned and repeatable fashion.
4.10	A system to ensure that any necessary inspections and testing is deligently carried out.
4.11	A system to ensure that key measuring equipment is properly maintained and calibrated.
4.12	A system to keep track of which material has been tested.
4.13	A system to prevent the inadvertent sale or use of nonconforming material or product.
4.14	A system to make sure that corrective action is taken whenever a quality problem is discovered and a system to try to prevent future quality problems from occurring.
4.15	A system to make sure that the right items get to the right place safely and on time.
4.16	A system to maintain and safeguard documents and records that relate to product quality.
4.17	A system that conducts periodic internal audits to verify the integrity of the quality system.
4.18	A system to ensure that employees have received the appropriate training fot their jobs.
4.19	A system to ensure that servicing is carried out (if required by the sales contract).
4.20	A system to ensure that statistical techniques are used where appropriate and are properly applied.

subsystem. Often, the auditor will sample documents and records from day-to-day operation. This may be supplemented by employee interviews and physical inspections. By cross-checking the results, a skilled auditor can quickly determine whether the implementation is effective. Finally, the auditor will look at any performance objectives that the organization set for itself when it designed the management system. The auditor will examine records to see if the system is achieving those objectives. At this point, the auditor may recommend that a certificate of compliance be issued. This process is repeated every three years, with less extensive surveillance audits occurring every six months to a year.

The ISO 9001 approach includes two important subtleties. The first is its stress on documenting the management system. Many experts state that ISO 9001 requires a supplier to "say what you are going to do, then do what you said you were going to do."[5] It follows that the design of each subsystem must be documented and operational records must be kept. While this obsession with

paperwork is a source of irritation to many firms, it is critical to cost-effective auditing. If the management system did not generate documents and records, the auditor would take far longer to determine whether the system actually worked. All major metastandard initiatives demand this type of documentation and record-keeping. In effect, they are designed to create a situation where it is easier to implement the subsystems than to fake a paper trail that can escape auditor detection.

The second subtlety concerns the pass/fail nature of the metastandard. A supplier fails an ISO 9001 audit if *any* of the twenty subsystems does not operate correctly. While external stakeholders may not know exactly how the supplier implements a given subsystem, they can be confident that all twenty subsystems exist and function to some acceptable degree. It is this completeness test that gives compliance its meaning, not only for ISO 9001, but for all management metastandards.

Where are Metastandards Being Used?

Until the arrival of ISO 9001, most managers had never encountered a metastandard. Nevertheless, the technology did not develop overnight. The idea of using management system design rules to achieve management accountability first appeared in 1970s regulations as a few isolated clauses about management. The history of metastandards can be followed through a series of environmental laws, product safety regulations, health care, scientific research, higher education, finance, food safety, and law enforcement guidelines.[6]

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This history is important for two reasons. First, it shows that most major metastandards have emerged since 1992. For the first time, small groups of experts can change management practice in thousands of firms. This rate of innovation dwarfs most (perhaps all) previous efforts at changing management practice. Second, the history shows that metastandards are emerging as an interconnected and interrelated network. Technical and legal specialists have been working along two broad, but closely related, development streams: a) standards and regulations for product quality and safety, and b) standards to help organizations prevent actions that could create civil or criminal liability.

Quality and Safety Standards

Concern for management standards dates at least to the US Department of Defense's MIL-Q-9858A Quality Program Requirements, issued in 1959. This regulation set design criteria for the quality assurance programs for defense contractors.[7] Over the following two decades, remarkably similar metastandard requirements appeared in standards for manufacturing aircraft, nuclear equipment, pharmaceuticals, pressure vessels, and medical devices.

In the 1980s, Japanese manufacturers established product quality as a competitive weapon. The Japanese recognized that complex products are more vulnerable to bad components and poor supplier practices. To minimize their vulnerabilities, they demanded more information about their suppliers' quality management systems. Large industrial purchasers from all nations began to audit suppliers (termed second party auditing) and even to write their own quality assurance metastandards. For example, Ford Motor Company introduced the Q1 Preferred Quality Award in 1981. Seven years later, Ford made Q1 a standard by announcing that suppliers had to meet it if they wanted to continue selling to Ford.

Proprietary standards and second-party audits, however, proved costly for both customers and the suppliers. Large suppliers found themselves hosting dozens of audits a year, each to a different buyer's standard. Industry associations (e.g., the American Bureau of Shipping and the American Gas Association) tried to help by issuing voluntary standards. Some of these standards called for association inspectors to conduct periodic factory visits. However, most were narrowly written for the industry and fell short of the generality demanded of metastandards. In 1986, the US Government introduced the Malcolm Baldrige National Quality Award. Although it was set up mainly as a competition, several firms, including Motorola and Xerox, have adopted its judging criteria as a form of metastandard.

In 1987, the International Organization for Standardization (ISO) in Geneva published the ISO 9000 family of standards and updated them in 1994. At the outset, the twelve European Community (EC) countries adopted ISO 9000 to replace their varied national factory certification standards. To sell these safety-sensitive products in Europe, a supplier must either: a) submit regular samples for testing, or b) hire an independent auditor to certify that its manufacturing plant conforms to the ISO 9001 metastandard.[8]

As exporters saw the value of compliance, other nations rushed to embrace the standard (see Table 2). A little later, large industrial purchasers in the US (e.g., DuPont, General Electric and Eastman Kodak) began to demand ISO registration from their suppliers. In turn, many suppliers pressured their sub-suppliers to adopt the standard. A few suppliers began to seek registration to gain a marketing advantage during the transition period when most suppliers would not yet be registered. These aggressive moves forced other firms to seek defensive registrations. A few suppliers viewed compliance as a defense in criminal and civil product liability lawsuits, while others considered ISO 9000 a convenient base for more extensive total quality systems.

By March 1995, registered sites around the world exceeded 95,000. By December 1995, registered sites in the US passed 8,100. Participants included ALCOA, Allen-Bradley, AT&T, Caterpillar, John Deere, Exxon, Federal Express, GE, Georgia Pacific, IBM, Motorola, NCR, Texas Instruments, 3M, Unisys and Xerox. In a recent survey of medium to large-scale US manufacturers, more than half expressed a strong desire to seek certification.[9] The standard has been applied to many different kinds of organization, including manufacturers, distribution services, consulting services, software developers, public utilities, and even a few financial and educational institutions.[10]

Table 2
Growth in ISO 9000 Registration Activity

Date	Registered Sites—Worldwide[a]	Registered Sites—US[b]
Jun 87	Initial publication	
Jan 90	N/A[c]	1
Jan 93	27,824	893
Oct 93	45,546	2,059
Jun 94	70,517	3,960
Mar 95	95,476	5,954
Dec 95	N/A	8,400

[a] Source: Mobil Europe, Inc., reported in *Quality Systems Update*, 4(3), p12.
[b] Source: Various issues of *Quality Systems Update*.
[c] No data available.

As ISO 9000 gained acceptance, industrial purchasers, associations and government agencies harmonized their older quality standards.[11] One of the most dramatic developments occurred in 1994, when the big three US auto companies announced QS9000. QS9000 embeds ISO 9001, along with requirements that address specific auto industry concerns. By early 1996, both Chrysler and General Motors instituted mandatory 1997 deadlines on third-party audits of *all* parts suppliers.

Metastandards require careful interpretation and auditor training. In 1988, there were no US registrars. By 1994, there were more than fifty, including such well-known firms such as AT&T, Underwriter's Laboratory, and KPMG, as well as subsidiaries of European registrars such as British Standards Institute and Det Norsk Veritas.

The explosive growth of ISO 9000 registration is mirrored by the evolution of the third-party auditing and registration industry. Metastandards require careful interpretation and auditor training. In 1988, there were no US registrars. By 1994, there were more than fifty, including such well-known firms such as AT&T, Underwriter's Laboratory, and KPMG, as well as subsidiaries of European registrars such as British Standards Institute and Det Norsk Veritas. Driven by the sudden growth, the compliance verification industry has been a perpetual work in progress.

Laws and Regulations

The quality management standards required that suppliers have adequate management systems to avoid defective outputs. Another group of standards and regulations sought to *prevent behavior* that might expose organizations and their officers to either civil or criminal liability.

Legal Obligations for Due Diligence

Civil and criminal penalties for violating federal laws have risen dramatically in recent years, particularly laws involving consumer fraud, workplace safety and environmental protection. Many of these laws make organizations, officers and managers strictly liable, meaning that they are accountable for acts committed by employees, even if those acts violate official policy. In fact, officers can be jailed and organizations can face crippling fines if they fail to exercise due diligence to prevent foreseeable crimes by employees.[12]

In the past, due diligence was a complex matter for courts and juries to decide. However, the US Federal Sentencing Commission issued new sentencing guidelines in 1992 that contain an explicit definition. The commentary to the Guidelines describes an "effective program to prevent and detect violations of law" that is essentially a metastandard for the prevention of corporate crime.[13]

If an organization follows the guidelines, it can a) probably prevent crime, or b) reduce its liability in the event that prevention fails. The commentary reflects a growing trend in regulation and law aimed at rewarding compliance efforts as well as punishing wrongdoers. It also explains why many firms are instituting voluntary management system standards to comply with the financial, environmental, and workplace regulations that are discussed in the following sections.

Accounting and Internal Controls

At first glance, financial auditing standards resemble metastandards. The analogy breaks down, however, because accounting disclosure standards deal with outputs, not management processes. A better analogy involves the internal financial control systems that companies install to avoid financial errors and prevent fiscal wrongdoing. In the past, company managers designed the internal controls. Now, directors and managers are under growing pressure to install more formal controls.[14]

In 1977, the Foreign Corrupt Practices Act required all public firms to have some (albeit unspecified) system of internal controls.[15] In 1987, the debacle in the savings and loan industry spurred the National Commission on Fraudulent Financial Reporting (the Treadway Commission) to urge that public companies strengthen those controls.[16] By 1993, one public company in four, and approximately 60 percent of the Fortune 500 companies, made statements about internal controls in their annual reports. However, the term internal controls continued to mean different things to different firms. In 1988, the Auditing Standards Board of the American Institute of Certified Public Accountants (AICPA) issued a Statement on Auditing Standards No. 55 to guide auditors in analyzing and documenting a client's internal controls. SAS 55 focused on the management of auditing, but it also contained a number of rules to guide control system design.[17]

In September 1992, the Committee of Sponsoring Organizations of the Treadway Commission (COSO) proposed a framework with detailed criteria for judging internal control systems. In 1993, the Framework was officially sanctioned by the AICPA in its Statement on Standards for Attestation Engagements No. 2 as a reasonable basis for assessing internal controls.[18] The COSO Framework's metastandard addresses not only management systems, but also human resource policies, customer relations and regulatory compliance. As such, it is arguably the most general metastandard of any current significance. If anything, it may be too broad and flexible to serve as the basis for formal certification. However, it is an important milestone in making companies more accountable to shareholders for the competence of internal management processes.

Environmental Regulation

The US Environmental Protection Agency (EPA) administers dozens of laws regulating the handling and disposal of wastes and toxic substances. Recently, the penalties for violating these laws have increased sharply.[19] In addition, the Federal Sentencing Commission has drafted special sentencing guidelines for environmental crimes that will further increase the penalties for managers and organizations.

These threats have motivated many firms to install voluntary environmental management systems. They have also spurred industry associations and standards-making bodies to develop metastandards. In 1988, the Chemical Manufacturers Association (CMA) introduced the Responsible Care program, with six Codes of Management Practice. The Codes list design rules for systems to manage distribution, pollution prevention, process safety, employee health and safety, community awareness and emergency response, and product stewardship. In its original form, the CMA's member companies were required to conduct annual self-evaluations against the Codes. In June 1993, the CMA board voted "to pursue compliance verification programs to improve the credibility of Responsible Care."[20] The chemical industry hopes that these independent audits will prove their due diligence and improve their public credibility on environmental issues.

In a far broader initiative, ISO has issued a draft of ISO 14000, an Environmental Management System metastandard.[21] Although it is not scheduled to be finalized until mid-1996, ISO 14000 is designed to harmonize with ISO 9000 and is already subject to third-party audits in both Europe and the United States.[8] Given the precedent of the CMA Responsible Care's auditing initiative, the

Federal Sentencing Guidelines lurking in the background, and likely pressure from Europe, it seems only a matter of time before many US firms subject their environmental management systems to external audits.

Workplace Safety

Penalties for noncompliance with US Occupational Safety and Health Administration (OSHA) regulations have risen with the penalties for other federal crimes. OSHA has been exploring the use of metastandards to encourage prevention instead of punishment. This trust began with OSHA's 1982 Voluntary Protection Program, which urged private firms to establish a comprehensive management system with active employee involvement to control the potential safety and health hazards of the site.[22] In 1992, OSHA introduced a metastandard, the Process Safety Management of Highly Hazardous Chemicals regulation, and external auditing. This regulation requires firms to install hazardous material management systems where OSHA tells you what to do, not how.[23] Industry experts expect that OSHA will continue to shift its regulatory focus in favor of metastandards over the coming years.

Why are Metastandards so Popular?

In one recent study, the combined labor cost and cash expense to prepare a medium-sized plant for ISO 9001 ranged from $50,000 to $1 million, with a typical cost of $250,000. The time varied from six months to two years, with a year being typical.[24] So why are firms willingly expending this effort? Most want the certificate.

In short, earning a metastandard certificate is attractive because it promises to be a cost-effective way to stay in a stakeholder's good graces or gain a place on a customer's bid list. In an increasingly competitive global economy, where customers and stakeholders have burgeoning choices and diminished time, this is apparently sufficient reason to propel the dramatic emergence and growth of metastandards.

When the ISO introduced ISO 9000, it hoped that the compliance certificates would reduce transaction costs in world trade. Before ISO 9000, customers either performed on-site audits of their suppliers, or risked bad quality in their purchases. Now, the supplier pays for the audit and customers can rely on the certificate to demonstrate supplier quality. In theory, a supplier can fax the certificate to customers all over the world and each will know that the supplier has all of the twenty subsystems in Table 1.

Other metastandards appeal to similar logic. In each case, an independently audited certificate reassures stakeholders that the organization is competent enough. The COSO framework tells investors that companies' internal financial controls are adequate. The OSHA Process Safety Management audit tells the government that a company's safety management system is adequate. ISO 14000 compliance shows that a company's environmental management system is sufficient.

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How Will Metastandards Affect Management Practice?

When a new technology first appears, its implications are poorly understood. It takes time and experience to reveal the hidden assumptions, bad side-effects, and subtle opportunities that the new technology will create. Since metastandards are still a work in progress, no one knows their long term

impact. This raises an important question: will metastandards change our beliefs about organizations and management practice? This section examines four places where metastandards could lead to fundamental changes in management practice. The scenarios are speculative and far from certain. Nor are they exhaustive. Nonetheless, they should provoke scholars, managers and policy makers to begin thinking about some of the long-term implications of this phenomenon.

Metastandards May Legitimize a Satisficing Model of the Firm
An overlooked axiom is that a standard can never describe best practice. No matter how sternly written, a standard can always be improved by ingenious innovators. At best, standards establish what is good enough. This fits perfectly with a concept that Herbert Simon introduced in the mid-1950s. He argued that human decision-makers lack the intellectual capacity to consider every possible option. Instead, they must make a cursory scan of their environment to isolate a few options that merit rigorous evaluation. Simon coined the term "satisficing" to describe this type of search.[25]

If, as Simon argued, satisficing dominates real-world decisions, then published certificates are ideal facilitators. One glance at a metastandard certificate establishes that a complex management system is good enough. This puts ISO 9001 in a new light; it is the first successful attempt to apply and validate a satisficing criteria to management practice. The global, market-driven success of ISO 9001 shows that satisficing purchasers are both numerous and they have similar satisficing criteria in the management of quality.

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It also follows that optimizing stakeholders can never be satisfied with certificates. A demanding consumer is not impressed by the Underwriters' Laboratory sticker and a strong environmentalist will never accept that pollution is within acceptable limits. Thus, there are two competing ways of thinking about every management system:

- A model of a management system that is good enough to reassure satisficing external stakeholders.
- A model of the better management system that managers and internal stakeholders aspire to create.

This distinction explains a lot of recent management history. Traditional philosophies of product quality stressed consistency with specifications (satisficing) while TQM philosophies demand continuous improvement (optimizing). Environmental regulations laws have become steadily more onerous as regulators confuse the goal of minimized pollution with companies' desire to know what is good enough. Needless to say, many managers are caught by the contradictions.

By recognizing that satisficing and optimizing are separate concepts, managers may find it easier to sort out conflicting notions of standards. If the distinction is fully applied, future managers may come to see their management systems from both perspectives simultaneously. Even as they chase their ideal management system, they will constantly monitor to ensure that it stays good enough to satisfy external stakeholders.

Management scholars have an important role to play in helping managers deal with this perspective. Scholars have the skills to investigate the implications of

managing against multiple criteria. They also have the theory-building skills to find newer and better definitions of good enough. Without such theory, well-meaning standard writers may blunder. With tens of thousands of companies around the world following the same document, such errors could be very costly indeed.

Metastandards May Politicize Management Principles
In the past, managers and scholars have assumed that management prescriptions stand or fall on their own merit. If companies try a new idea and prosper, the idea spreads. If they try it and fail, the idea dies. A long history of management fads reinforces this view.

With metastandards, the situation is radically changed. When a prescription is written into a metastandard, it is quickly embedded in the prevailing stakeholder definition of good enough. Once entrenched, a metastandard will be hard to remove. This raises an obvious scenario where management innovations are driven not by empirical experience, but by strong political processes that seek to control the conceptual content of the published standards. Political pressures are likely, because the emerging metastandards have many requirements that affect different realizations within the organization. For example, most require that the management system be formally documented, the authority and responsibility be formally defined, and that senior management take clear ownership of the system. Most also require a formal system of internal auditing, a formal training system, and a formal system to manage change. Some go so far as to require formal systems of employee participation.

At worst, these overlapping metastandards could marginalize a great deal of management research. Scholarly results would not be wrong, just irrelevant. The battleground of new management ideas would shift from the marketplace to the committee rooms, where standards-writing experts debate the next edition. Powerful stakeholders would jockey for position like political parties, based as much on ideology as science. Scholars would get to study the results, after they had influenced tens or even hundreds of thousands of organizations around the world.

At best, those same political processes would amplify the consensus positions of experts from around the world. Rather than hurting organizations, metastandards would propel an accelerated implementation of sound management practice over vast stretches of the organizational landscape. It is even possible that, as standards writing bodies gain stature and resources, they will call upon scholars to validate ideas before casting them in stone.

Metastandards May Lower Organizational Diversity and Slow Management Innovation—Or Greatly Accelerate Both
Experience with technical standards suggests that standards, once adopted, are very hard to repeal. Switching costs deter users from adopting a new and better standard. This is true even when the standard has objectively outlived its usefulness.[26] Thus, there is always a risk that widespread application of metastandards will stifle management innovation.

In a worst case scenario, metastandards will increase the personal risks for managers who want to try new ideas. Decision makers might require that manager X's new idea for structuring worker empowerment be approved in

advance by the COSO (or ISO 9000 or OSHA) auditor. Faced with this barrier, decision makers may avoid experimentation. Worse, the standards may have a chilling effect even on normal, incremental innovation that forms the bulk of management progress. If companies fear for their certificates, they may discourage the kind of routine experimentation and risk-taking that is central to progress.

On the other hand, management system metastandards might create a more optimistic scenario. Once chosen for the IBM PC, Microsoft DOS became the de facto world standard for microcomputer software development. Switching costs kept competing operating systems at bay for over a decade. Despite this monopoly, DOS fostered extraordinary innovation. It did deter operating system innovations, but its huge market share attracted a cornucopia of hardware and software innovations at the application level. As a result, the IBM PC industry has seen an astonishing technological development, some of it even spurred by the need to work around the constraints of DOS.

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A similar scenario could occur with metastandards. Innovations that fit *with* the metastandard should hold great interest for the large numbers of firms that meet the standard. This compliance will mean that they will all have similar needs. This, in turn, could create huge markets for certain types of management innovation, and these innovations will spread much faster than in the past. There are already indications of this in the growing use of corporate intranets and groupware. All of the major metastandards demand formal document control systems that are ideally suited for electronic implementation. Consequently, companies like IBM (which purchased Lotus Notes) and Interleaf (which has announced an ISO 9000 support initiative) have begun to see the outline of a vast new market.

It remains to be seen how the two scenarios will play out. Will the future be a world where overlapping regulations and standards thoroughly stifle management innovation? Will the future be a world where metastandards provide a critical mass of common management structure that fosters a massive new market for management innovation? Managers, scholars, and policy makers should be sensitive to the implications of both possibilities.

Metastandards May Give Stakeholders Much Greater Control

Whatever else may emerge from metastandard technology, it seems certain that external stakeholders will gain more control over internal organizational activity. A preliminary review of the ISO 9000 phenomenon suggests that there are at least four ways that this could occur:

- Control of information: Compliance certificates provide valuable information about general competence. If competing firms have certificates, a company's failure to obtain certification may cause concern. A change in compliance status may trigger even more concern. It may be far worse to fail the ISO 9000 audit than to pass it and then not maintain it. A certificate suspension would send a clear and damaging message to stakeholders.
- Influence through the standards writers: Powerful stakeholders can lobby standards-writers to enforce a specific satisficing model on large numbers of firms. When ISO 9001 was written by a small group of experts in the mid-1980s, few managers or scholars even knew of its existence. Now major multinational corporations are demanding a role in the standards-writing

process.[27] By demanding that all parts suppliers *must* achieve audited compliance with QS9000 (a superset of ISO 9000) GM and Chrysler are effectively imposing their quality management system on thousands of independent manufacturing firms.

- Influence through the auditor: With the publication of metastandards, stakeholders can direct complaints to the auditors as well as to the organization. The precedent for this behavior is already established in the accounting industry. Stakeholders can also give greater weight to certificates from auditors who are tough than to auditors they suspect of being lenient. This, in turn, will encourage good companies to seek certification from tough auditors and marginal companies to seek out the easy touch. Even with rigorous accreditation of auditors (a condition that does not yet exist for many metastandards), variations may persist.
- Control through the courts: As a last resort, matters can be decided by law. Several legal scholars have raised concerns about metastandards and liability. The American Bar Association has suggested that the COSO Framework may raise stakeholders' expectations. Meeting the standard may strengthen a firm's defense, while failure to meet it may expose the firm to even greater liability.[28] To compound the problem, metastandard standards often refer to other standards. There is a strong connection between management standards and criminal liability for corporate officers. For example, the Federal Sentencing Guidelines is especially stern if there is evidence that an organization has failed to meet applicable standards. It remains to be seen how much of metastandard compliance will be considered applicable.[29]

These mechanisms presage a management future with many unknowns. On one hand, metastandards may force companies to manage their operations far more defensively. Alternatively, companies may recognize that stakeholder control is limited by the metastandard's definition of good enough. By exceeding the metastandard, a company may actually shake off some of a stakeholder's current influence. Only time will tell.

A New World of Management Standardization?

Management system metastandards are not simply another management fad. They represent a new management technology that may have a profound impact on the near-term evolution of management practice. It is now clear, for example, that metastandards are technically feasible, that they can affect management theory and practice, and that they have the potential to fundamentally change the power relationship between organizations and key stakeholders.

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However, there is no compelling evidence that metastandards are ultimately good or bad. Nevertheless, the rapid acceptance of ISO 9000 and COSO by US companies suggests that many firms find that the standards are well written and intrinsically worth observing. For regulated companies, the ability to design their own management systems within the rules of a metastandard may even provide a relief from the instruction manual approach to government regulation. At the same time, metastandards are behaving very differently from previous management innovations. They are not seeping slowly into practice. Instead, powerful stakeholders are aggressively championing their immediate use. Entire compliance industries are springing up, populated by firms with a vested interest in spreading the gospel. As the standards spread and evolve, stakeholders and standards-making bodies will draw policy makers, managers, and scholars farther into uncharted territory. With opportunities and threats of

this magnitude, it is the author's hope that scholars, practitioners, and policy-makers will begin the serious task of mapping and analyzing the broader implications.

Endnotes

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[4] The characterizations in this table were strongly influenced by descriptions in: Peach, R. (ed), 1994, *The ISO 9000 Handbook*, 2nd ed, CEEM Information Services, Fairfax, VA.

[5] This implicitly assumes that the firm says it will implement all of the subsystems in the ISO 9001 model. See: Rick Mullin, 1993, General registers Green River site: first ISO 9002 for natural soda ash, *Chemical Week*, 152:59 Apr 28, 1993; J.D. Flister, and J.J. Jozaitis, PPG's journey to ISO 9000, *Management Accounting*, July 1992, 74:33-5+; C. Cirulli, Adventures in the ISO jungle, *The Journal of European Business*, March/April, 23-30, 1993.

[6] For other MSM examples not discussed here, see: a) National Advisory Committee on Microbiological Criteria for Foods, Hazard Analysis and Critical Control Point System, *International Journal of Food Microbiology*, v16, 1992, pp. 1-23; b) DeWoskin, Robert S., Good Laboratory Practice Regulations, in *Principles of Research Data Audit*, A. Shamoo (ed), Gordon and Breach, New York, NY, 1989; and c) Joint Commission on Accreditation of Healthcare Organizations, *The Joint Commission Guide to Quality Assurance*, Joint Commission on Accreditation of Healthcare Organizations, Chicago IL, 1988, pp. 13-14.

[7] US Defense Supply Agency, 1991, *Defense in-plant quality assurance program*, Department of Defense, Defense Supply Agency, Alexandria, Virginia.

[8] ISO has more than 150 member nations. It sponsors and publishes standards that promise to facilitate international trade, but leaves enforcement to the nations and organizations that adopt the standards. ISO's standards are developed by technical committees staffed by volunteer experts from around the world. ISO 9000 was developed by technical committee TC176, which based its work on a British standard (BS5750), that was descended from the venerable MIL-Q-9858A. See Peach, Op. Cit. p. 8.

[9] From a survey conducted in the summer of 1993 for the National Association of Manufacturers. See Swamidass, P.M., *Technology on the factory floor II: Benchmarking manufacturing technology use in the United States*, The Manufacturing Institute, Washington DC, December 1994.

[10] For applications in software development see: Brown, J., Developers weigh necessity for ISO 9000 certification, *Computer Reseller News*, Aug 29, 1994, n593, p.101, 104. For developments in electric utilities, see: Bretz, E.A., Power producers consider ISO's quality stamp of approval, *Electrical World*, Nov 1994, v208n11, p. 82-84. For applications in European education, see: Solomon, H., Total quality in higher education, *Management Services*, Oct 1993, v37n10, p. 10-15.

[11] In November 1993, the FDA proposed rewriting the Good Manufacturing Practices regulation to match ISO 9001 in: US Food and Drug Administration, 1993, *Medical Devices; Current Good Manufacturing Practice (CGMP) Regulations; Proposed Revisions*, published in 58 Federal Register 224, November 23, 1993. In December 1993, Boeing Commercial Airplane Group adopted ISO 9002 as the company's basis for meeting FAA requirements (*Quality Systems Update*, 4(3) Mar 1994:15). Early in 1994, the Department of Defense and NASA issued letters stating that DoD's MIL-Q-9858A and NASA's 5300.4(1B) quality assurance standards would be replaced by ISO 9001 (*Quality Systems Update*, 4(3) Mar 1994:1). The American Iron and Steel Institute introduced a voluntary program in 1990, in which third-party assessors would audit participating suppliers against ISO 9002 (Peach, 1994:473). In late 1991, the Railway Interest Group of the National Association of Purchasing Managers recommended that railways use ISO 9002 for supplier certifications (Railway Age, Jan 1992:18). Chrysler, Ford and General Motors announced late in 1994 that their first-tier suppliers would have to comply with ISO 9001 (*Quality Systems Update*, 4(4) April 1994:1).

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